

JACC STATE-OF-THE-ART REVIEW

Cardiogenic Shock Survivorship

A Multidisciplinary Approach Across the Continuum of Care

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ABSTRACT

Cardiogenic shock (CS) is a complex and heterogeneous clinical syndrome associated with high short-term mortality as well as a long-term burden of morbidity. Whereas short-term survival has improved owing to early recognition, revascularization, and temporary mechanical circulatory support (tMCS), survival after hospital discharge remains poorly defined. A large number of CS survivors experience persistent myocardial dysfunction, neurocognitive impairment, psychological distress, frailty, and recurrent heart failure hospitalizations. Yet current clinical guidelines and research frameworks provide limited direction beyond the acute phase, underscoring the urgent need for a structured longitudinal approach to care. This State-of-the-Art Review proposes the Cardiogenic Shock Survivorship Continuum, a novel framework delineating 3 interdependent phases: acute rescue and stabilization, assessment and optimization of in-hospital trajectories, and postdischarge shock care. The acute phase focuses on timely diagnosis, end-organ support, and individualized tMCS strategies. The in-hospital phase emphasizes diagnostic reassessment, heart failure guideline-directed medical therapy initiation, and procedural interventions targeting reversible pathophysiology. The postdischarge phase advocates for structured outpatient models, multidisciplinary postshock clinics, individualized risk stratification, re-referral for advanced therapies when appropriate, and recovery-focused rehabilitation. This review underscores the need to redefine success in CS, identifies critical gaps in evidence, and proposes future directions in research, clinical infrastructure, and collaborative networks to advance the field. Optimizing care in CS requires a paradigm that emphasizes both short- and long-term survival, with emphasis on achieving functional recovery, stability in health status, and preservation of quality of life. (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

Cardiogenic shock (CS) is a complex and heterogeneous clinical syndrome associated with high mortality despite advances in acute-phase care. Contemporary improvements in early recognition, targeted revascularization, and temporary mechanical circulatory support (tMCS)

have increased in-hospital survival to nearly 60% to 70%.¹⁻⁴ Long-term outcomes, however, remain sobering. One-year mortality exceeds 40% to 60%, and nearly 1 in 5 patients are rehospitalized within 30 days.⁵ Moreover, postdischarge morbidity remains largely uncharacterized, with emerging data showing

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**ABBREVIATIONS
AND ACRONYMS****AMI** = acute myocardial infarction**CS** = cardiogenic shock**dmCS** = durable mechanical circulatory support**ECMO** = extracorporeal membrane oxygenation**GDMT** = guideline-directed medical therapy**HF** = heart failure**HRQoL** = health-related quality of life**LVAD** = left ventricular assist device**PICS** = post-intensive care syndrome**PRO** = patient-reported outcome**tMCS** = temporary mechanical circulatory support**VA** = venoarterial

that survivors face substantial multisystem complications, neurocognitive and psychologic sequelae, and frailty.^{6,7} Although inpatient heart failure (HF) guideline-directed medical therapy (GDMT) initiation and use of tMCS may improve early survival,^{8,9} post-discharge care models have not aligned with the growing population and multifactorial considerations for CS survivors. Existing literature provides limited direction beyond the acute hospitalization, and clinical trials have focused almost exclusively on short-term outcomes, rarely incorporating postdischarge survival, functional recovery, or health-related quality of life (HRQoL) as relevant endpoints.¹⁰⁻¹²

Survivorship in CS encompasses more than survival to discharge; it reflects a continuum of vulnerabilities such as persistent myocardial dysfunction, multiorgan impairment, neurocognitive and psychologic sequelae, frailty, functional decline, critical illness myopathy and neuropathy, recurrent

HF hospitalizations, and late mortality,^{6,7} closely paralleling the construct of post-intensive care syndrome (PICS).¹³ Despite this burden, the field lacks structured postacute care models, standardized transition-of-care strategies, and consensus regarding recovery milestones. Moreover, the terminology used to describe postshock trajectories (eg, recovery, remission, native heart survival¹⁴), remains inconsistent across studies. Without definitional precision, benchmarking outcomes, aligning therapeutic goals, and designing clinical trials that capture the continuum of CS care to close key evidence gaps become exceedingly difficult, limiting progress across care delivery and research.

To address these gaps, this State-of-the-Art Review introduces the Cardiogenic Shock Survivorship Continuum, a framework that reconceptualizes CS not as an episode ending at discharge, but as a dynamic, longitudinal, multidisciplinary process. The model outlines 3 interconnected phases of CS care: 1) acute rescue and stabilization, focused on early CS identification, rapid hemodynamic stabilization, prevention of further deterioration, and restoration of end-organ perfusion; 2) in-hospital optimization and trajectory assessment, a transitional phase in which patients can achieve hemodynamic stability but remain biologically vulnerable, creating a critical window for targeted diagnostics, HF GDMT initiation when feasible, revascularization or structural/electrophysiologic interventions, and tMCS weaning; and 3) postdischarge shock care, extending into the

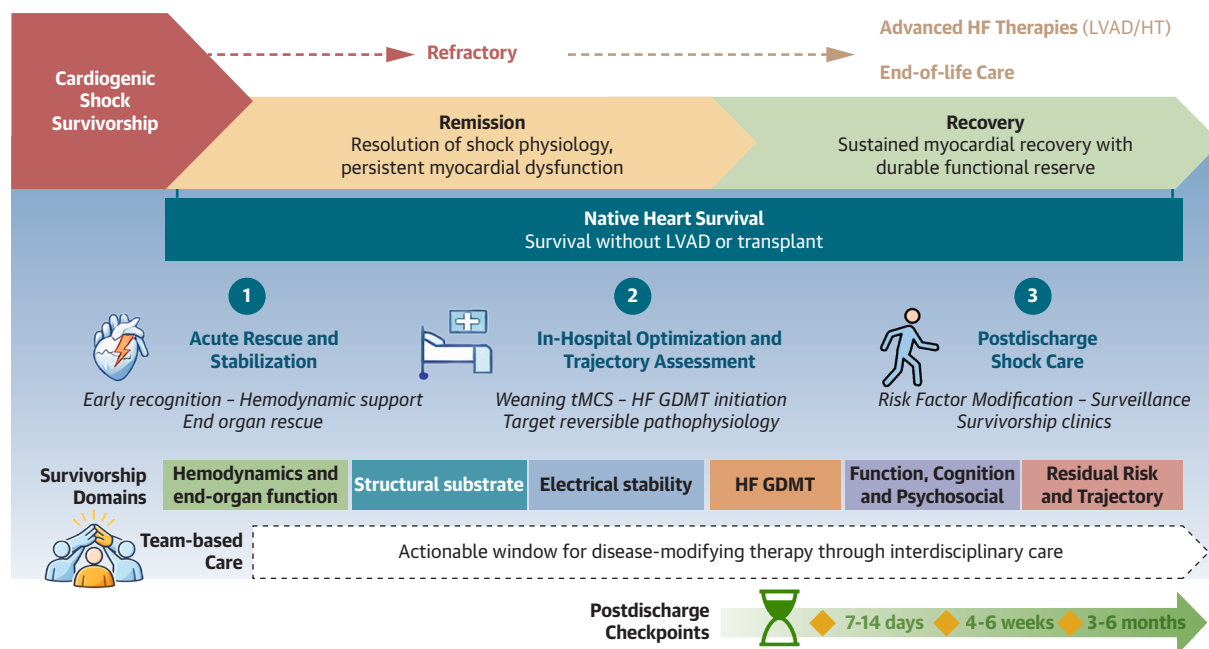
outpatient setting, where the goals are to minimize rehospitalizations, advance medical therapy, address physical and psychosocial sequelae, and refine candidacy for advanced HF therapies. Together, these phases reframe CS survivorship as a dynamic continuum, providing a clinically actionable structure for care pathways, resource planning, and future research.

**TOWARD A STANDARDIZED DEFINITION OF
SURVIVORSHIP TRAJECTORIES**

The nomenclature used to describe post-CS trajectories remains imprecise, with terms such as recovery, remission, and native heart survival applied variably and sometimes interchangeably. This semantic ambiguity mirrors early challenges in oncology before the adoption of standardized response frameworks and hampers consistent communication, prognostication, therapeutic alignment, and clinical trial design.^{14,15} Oncology response frameworks such as Response Evaluation Criteria in Solid Tumors (RECIST)¹⁵ formalize complete response, partial response, stable disease, and progressive disease as distinct biological states while explicitly recognizing that failure to achieve response or disease control leads to progression, treatment intensification, or transition to non-curative pathways. Applying a similar response-based construct to CS clarifies which patients recover, which enter remission, and which progress toward requiring advanced therapies or end-of-life care (**Central Illustration**).

As such, drawing from established constructs in oncology, HF, critical care, and PICS literature, we propose a response-based lexicon that conceptualizes postshock states as dynamic survivorship trajectories rather than fixed CS states (**Figure 1**)¹⁴:

- **Recovery:** Recovery represents the analogue of a complete response. It reflects true myocardial recovery characterized by sustained reverse remodeling, normalization of biventricular function, restoration of molecular and cellular homeostasis, and durable improvement in functional capacity and HRQoL.^{16,17} Recovery implies biological resilience and a low likelihood of relapse under physiologic stress.
- **Remission:** Remission is the most frequent post-CS trajectory and parallels a partial response or stable disease in oncology. It is defined by resolution of shock physiology and stabilization of systemic perfusion with partial, but incomplete, reversal of the HF phenotype. Patients in remission exhibit persistent myocardial abnormalities, limited

CENTRAL ILLUSTRATION Cardiogenic Shock Survivorship Continuum

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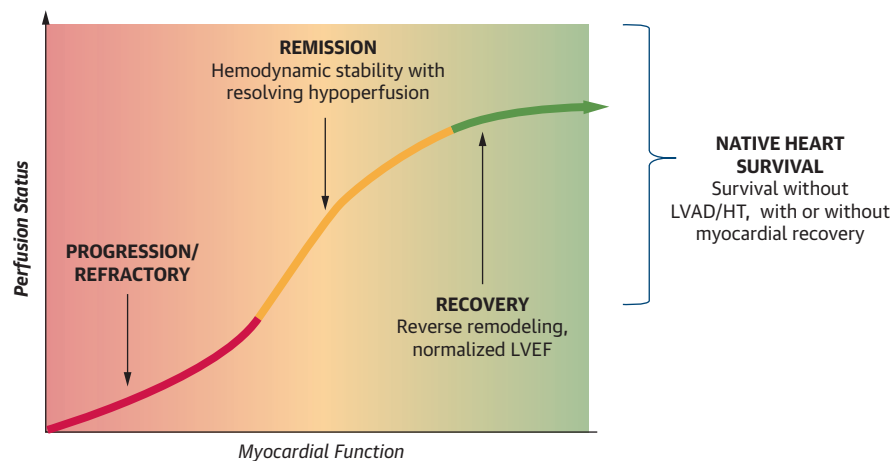
Following the acute shock state, patients transition along response-based survivorship trajectories, including recovery and remission, with failure to achieve durable stabilization resulting in a refractory trajectory requiring advanced heart failure therapies or end-of-life care. These survivorship constructs are not mutually exclusive; overlap is expected, and patients may move between trajectories as physiology evolves, reflecting clinical reality and guiding, rather than constraining, decision making. Interdisciplinary care across the cardiogenic shock continuum is organized into 3 overlapping phases: 1) acute rescue and stabilization, focused on early recognition, hemodynamic support, and end-organ rescue; 2) in-hospital optimization and trajectory assessment, a biologically vulnerable phase representing a critical, actionable window for disease-modifying therapy, multidisciplinary evaluation, guideline-directed medical therapy (GDMT) initiation, and weaning of temporary mechanical circulatory support (tMCS); and 3) postshock discharge care, emphasizing surveillance, relapse prevention, and survivorship-focused follow-up. This framework emphasizes the multisystem nature of cardiogenic shock survivorship, integrating cardiac and noncardiac organ function, functional status, cognition, psychosocial health, and evolving trajectory risk over time. HF = heart failure; HT = heart transplant; LVAD = left ventricular assist device.

physiologic reserve, and heightened susceptibility to recurrent decompensation and multiorgan sequelae, reflecting plastic rather than elastic remodeling.¹⁸⁻²⁰ Remission is therefore a state of disease control, not cure, and requires ongoing surveillance and active management.

- **Native heart survival:** Native heart survival is a pragmatic survivorship construct analogous to event-free survival in oncology. It refers to survival without durable mechanical circulatory support (dMCS) (left ventricular assist device [LVAD]) or heart transplantation, regardless of residual ventricular dysfunction, functional limitations, or degree of remission. This construct captures an important, heterogeneous population whose

survival may coexist with substantial ongoing myocardial, systemic, and psychosocial burden.

Throughout this review, we adopted this refined lexicon as the foundation for the Cardiogenic Shock Survivorship Continuum. Importantly, these survivorship constructs are not mutually exclusive. Overlap is expected, and patients may move between these categories as their physiology evolves. Acknowledging this fluidity reflects clinical reality and ensures that terminology guides, not constrains, clinical decision making. Despite this overlap, distinguishing these trajectories has practical value in that they help align expectations for prognosis, standardize outcome reporting, and inform the design of postshock care pathways and clinical trials.¹⁹⁻²¹

FIGURE 1 Clinical and Biological States After Cardiogenic Shock: Defining Remission, Recovery, and Native Heart Survival

The cardiogenic shock continuum encompasses distinct postshock states defined by tissue perfusion and myocardial function. Cardiogenic shock remission denotes resolution of shock and reversal of end-organ dysfunction with hemodynamic and clinical stabilization, without necessarily achieving normalization of myocardial structure or function. Myocardial recovery represents a biological and functional state characterized by reverse remodeling and restoration of normal or near-normal left ventricular ejection fraction (LVEF), permitting independence from advanced heart failure therapies. "Native heart survival" describes sustained compensated cardiac function, either with myocardial recovery or with persistent myocardial impairment, without the need for durable mechanical circulatory support (left ventricular assist device [LVAD]) or heart transplantation (HT). Differentiating these states is critical for accurate patient phenotyping, risk stratification, and the development of targeted longitudinal management strategies.

CARDIOGENIC SHOCK SURVIVORSHIP CONTINUUM: A NOVEL FRAMEWORK DELINEATING 3 INTERDEPENDENT PHASES

ACUTE RESCUE AND STABILIZATION. The acute phase of CS represents a decisive inflection point, in which hemodynamic instability and end-organ dysfunction converge to drive early mortality and shape downstream morbidity. The existing literature has extensively addressed the pathophysiology, diagnostic algorithms, and acute interventions in CS,^{22,23} so this section highlights those elements most relevant to survival: early recognition, timely escalation and de-escalation of tMCS, and mitigation of early physical and psychosocial stressors that frequently determine postdischarge disability.

Early recognition and risk stratification. Timely recognition and phenotyping of CS is essential to optimize chances of survival and functional recovery. CS is frequently underrecognized in non-acute myocardial infarction (nonAMI-CS) contexts, including decompensated HF, valvular emergencies, and myocarditis,²⁴⁻²⁶ because these presentations frequently lack hypotension or classic hemodynamic collapse until late in the disease course. Ranging from stage A ("at risk") to stage E ("extremis"), the

Society for Cardiovascular Angiography and Interventions (SCAI) stages provide a dynamic prognostic framework for stratifying patients based on severity, trajectory, and risk of deterioration.^{25,27} Studies show that most patients move quickly between SCAI stages within 24 hours of diagnosis, especially stage B patients, who have a high likelihood of rapid deterioration if not aggressively managed.²⁸ In fact, recent emerging data suggest that clinical deterioration within the first 6 hours may be associated with short-term mortality.²⁹ These observations underscore the critical importance of early recognition and serial reassessment, not only for accurate prognostication, but also for identification of patients who may benefit from shock team activation and escalation of care.³⁰

A central challenge in CS is trajectory-based risk stratification, specifically, distinguishing patients with salvageable physiology and native heart survival potential from those in whom ongoing escalation of support is futile or unlikely to change outcome. Traditional risk scores, such as CardShock, IABP-Shock II, and the Inova Heart and Vascular Institute risk score, offer important prognostic insights related to in-hospital mortality but fail to differentiate between reversible and irreversible

myocardial injury.³¹ These models rely on variables assessed at the time of presentation and do not capture dynamic changes in physiology, imaging, or biomarkers that may emerge over time. Consequently, some patients are maintained on advanced support beyond the window of meaningful recovery, while others may be prematurely denied escalation. Recent literature underscores the need for multidimensional, real-time prognostic tools that integrate physiologic, biochemical, and imaging data to better inform decision-making.¹⁶ Emerging evidence supports the incorporation of clinical features such as duration of shock, need for multi-inotropic support, and the presence of multiorgan dysfunction, which correlate strongly with lower likelihood of survival.^{3,32}

tMCS: escalation and de-escalation strategies. Use of tMCS has fundamentally reshaped the management of CS by restoring systemic perfusion, attenuating end-organ injury, and creating a pseudophysiologic interval during which etiology, reversibility, and destination therapy can be clarified. From a survivorship perspective, tMCS should be conceptualized not solely as rescue therapy, but also as a time-dependent intervention that influences downstream functional trajectory, including myocardial unloading, end-organ recovery, feasibility of rehabilitation, and candidacy for dMCS or transplantation. Contemporary platforms range from intra-aortic balloon pump to percutaneous microaxial LVADs (Impella CP/5.5) and venoarterial extracorporeal membrane oxygenation (VA-ECMO), with increasing adoption of combinatorial strategies (eg, ECMO with left ventricular unloading) in patients with profound circulatory collapse, biventricular failure, or concomitant respiratory collapse.

Escalation of tMCS requires integration of shock severity and trajectory, shock phenotype, and anticipated reversibility.² Although SCAI staging provides a validated framework for severity stratification, it is not prescriptive for device selection, escalation thresholds, or timing, which remain heterogeneous across centers.^{27,33} Observational data suggest that earlier support initiation, before irreversible end-organ dysfunction, may improve short-term survival,³⁴ but prospective phenotype-guided validation remains limited. Practical escalation decisions therefore depend on dynamic assessment of perfusion adequacy and shock progression, including lactate kinetics, vasoactive requirements, invasive hemodynamic profiles, and the systemic inflammatory response. Importantly, response to tMCS is not uniform and reflects both the myocardial substrate (acute ischemia vs chronic HF, restrictive physiology,

right ventricular [RV] dominance) and host factors (comorbidity burden, frailty, renal/hepatic vulnerability), underscoring the need for longitudinal reassessment rather than static “device-first” algorithms.

De-escalation is equally central to survivorship and should be approached as a milestone-based strategy aimed at minimizing device-related complications, enabling mobilization, and shifting toward durable stability, whether via native heart survival, durable LVAD, transplantation, or palliative goals of care. Successful de-escalation requires objective evidence of resolved shock physiology and sustainable cardiac performance, supported by invasive hemodynamics and perfusion metrics. Key physiologic targets include adequate forward flow (cardiac index >2.2 L/min/m²), restoration of ventricular pulsatility and RV reserve (eg, pulmonary artery pulsatility index >1.5 in RV-dominant shock), and normalization of right- and left-side filling pressures with improving end-organ markers (lactate clearance, mixed venous oxygen saturation).³⁵ Beyond physiologic thresholds, procedural and access decisions meaningfully influence survivorship: Axillary or subclavian strategies facilitate early ambulation and rehabilitation, whereas femoral cannulation is associated with immobility, vascular complications, and deconditioning.³⁶ Thus, escalation and de-escalation should be framed as an integrated survivorship pathway in which device selection, access approach, and weaning strategy are aligned not only with survival to discharge, but also with the probability of durable stability and functional reintegration.

Physical, psychological, and social stressors. From a survivorship perspective, the acute phase should be viewed not only as hemodynamic rescue, but also as the first opportunity to shape downstream disability and rehospitalization risk. Survivors of CS frequently manifest early elements of PICS,⁶ including delirium, neuromuscular weakness, deconditioning, depression, anxiety, and evolving social and financial stressors, each of which can impair adherence, follow-up engagement, and reintegration into daily function. Delirium and intensive care unit-acquired weakness are common in cardiac ICU populations and are independently associated with prolonged hospitalization, functional decline, and late mortality.^{37,38} These risks may be amplified in patients requiring prolonged vasopressors, sedation, and tMCS. Despite increasing awareness, early mobilization and rehabilitation remain inconsistently implemented in CS, often because of perceived hemodynamic instability, cannulation constraints, and staffing limitations.³⁹ In selected patients, particularly those supported through ambulation-

TABLE 1 Coronary Revascularization Strategy in AMI-Related CS

Study	n	Population	Study Type	Intervention	Key Finding
Hochman et al, 1999 ^{12,41} (SHOCK)	302	AMI-CS	RCT	Early PCI/CABG vs delayed	Early revascularization improved survival (short- and long-term)
Thiele et al, 2017 ^{4,40} (CULPRIT-SHOCK)	706	AMI-CS + multivessel CAD	RCT	Culprit-only vs immediate multivessel PCI	Culprit-only reduced 30-day death and RRT vs immediate multivessel PCI
Gill et al, 2022 ⁴²	75,431 (16 studies)	AMI-CS	Systematic review/meta-analysis (mostly observational)	Multivessel vs culprit-only PCI	Multivessel PCI associated with worse early outcomes; no clear long-term benefit
Lemor et al, 2020 ⁴²	198	AMI-CS (protocolized early MCS; MV-CAD subset)	Prospective cohort	Multivessel vs culprit-only PCI	Similar survival and AKI: suggests selected patients may tolerate multivessel PCI under protocolized care
Choi et al, 2023 ⁴⁴	315	AMI-CS on VA-ECMO	Multicenter cohort	Multivessel vs culprit-only PCI	Results vary by timing/strategy: highlights heterogeneity in ECMO-supported cohorts.

Tables 1 to 4 summarize representative studies informing interventional strategies during the postrescue in-hospital phase of CS and are intended to provide a curated, high-yield synthesis rather than an exhaustive inventory of the literature. Because evidence across several domains is derived predominantly from observational registries, administrative data sets, and small mechanistic cohorts, interpretation requires caution given heterogeneity in shock phenotypes, MCS strategies, illness severity, and procedural timing, as well as confounding by indication and survival and immortal-time biases. Accordingly, the article text provides additional context regarding patient selection, feasibility constraints, generalizability, and operational considerations for clinical application, while explicitly highlighting methodologic limitations and key evidence gaps that should be prioritized in prospective studies. SCAI shock stage was not uniformly reported across studies, particularly in earlier registries and observational cohorts. As such, no assumptions regarding shock severity were made for studies lacking formal classification.

AKI = acute kidney injury; AMI = acute myocardial infarction; CABG = coronary artery bypass grafting; CAD = coronary artery disease; CS = cardiogenic shock; ECMO = extracorporeal membrane oxygenation; MCS = mechanical circulatory support; MV = multivessel; PCI = percutaneous coronary intervention; RCT = randomized controlled trial; RRT = renal replacement therapy; SCAI = Society for Cardiovascular Angiography and Interventions; VA = venoarterial.

permissive strategies, rehabilitation during the index hospitalization is feasible and may reduce downstream disability.²⁴

Overall, defining and validating short-interval trajectory metrics (eg, 6-hour deterioration, failure of lactate clearance, or persistent congestion despite support) could enable enrichment strategies and more interpretable assessments of acute interventions while mitigating survivorship bias. Similarly, procedural decisions during rescue, including cannulation strategy and the feasibility of ambulation, function as survivorship interventions with potential downstream effects on ICU-acquired weakness and PICS. Collectively, acute rescue decisions determine not only whether the patient survives hospitalization, but also whether the patient leaves the ICU on a trajectory toward durable stabilization or toward recurrent decompensation, functional decline, and rehospitalization, highlighting the need for structured inpatient transition pathways and operational postshock longitudinal care models.

IN-HOSPITAL OPTIMIZATION AND TRAJECTORY ASSESSMENT. Following initial rescue, patients enter a high-stakes inpatient interval of optimization and trajectory assessment, in which transient stabilization occurs despite ongoing myocardial dysfunction and systemic injury. During this phase, patients remain vulnerable to recurrent hypoperfusion, persistent congestion, arrhythmic instability, ongoing end-organ injury, and device-related thrombotic and bleeding complications, with high near-term rehospitalization risk despite apparent

clinical stabilization. Accordingly, this interval demands integrated and multidisciplinary reassessment, combining invasive hemodynamics, biomarkers, imaging, and end-organ recovery, to define prognosis, identify reversible contributors, and guide destination planning toward native heart survival vs dMCS, transplantation, or comfort-focused pathways.

Multidisciplinary strategies to modify trajectory: targeting reversible pathophysiology. This phase provides a pivotal opportunity to shift from hemodynamic rescue to trajectory-modifying and recovery-oriented management. This period provides a unique therapeutic window for targeted interventions, including revascularization, electrophysiologic therapies, and structural heart procedures, that can modify trajectory and improve the likelihood of long-term stabilization (Tables 1 to 4). Importantly, a multidisciplinary approach is essential to identifying and treating reversible contributors to myocardial injury (Figure 2), with the goal of optimizing the probability of native heart survival and minimizing the need for escalation to dMCS or transplantation.

Revascularization. Early revascularization improves outcomes in AMI-related CS: Among patients with multivessel coronary disease, randomized data support an initial culprit-only percutaneous coronary intervention strategy. In contrast, observational data sets, including cohorts supported with VA-ECMO, highlight substantial heterogeneity in contemporary practice and remain insufficient to

TABLE 2 Structural Interventions in CS

Study	n	Population	Study Type	Intervention	Key Finding
Mitral TEER in cardiogenic shock with severe MR					
Simard et al, 2022 ⁴⁴	3797	CS + significant MR	Multicenter registry	TEER (MitraClip)	High feasibility; MR reduction associated with lower 1-y mortality
Tang et al, 2021 ⁵⁰	622 of ~38,000	CS + MR	Claims-based comparative	TEER vs medical therapy	TEER associated with lower in-hospital and 1-y mortality in adjusted analyses
Falasconi et al, 2021 ⁵¹	31	Refractory CS + severe MR	Multicenter registry	TEER	Feasible in refractory CS; procedural success associated with survival
Dimitriadis et al, 2025 ⁵²	~1,200 (14 studies)	CS + MR (pooled studies)	Systematic review/meta-analysis	TEER	Pooled mortality estimates support feasibility in selected patients
Severe AS with shock: TAVR/BAV strategies					
Kolte et al, 2018 ⁵³	40,042 (3,952 urgent)	Severe AS ± shock	TVT registry	TAVR	Urgent/emergent TAVR feasible; shock associated with higher early mortality
Frerker et al, 2016 ⁵⁴	27	AS-CS	Single-center series	Emergency TAVR	Rescue TAVR feasible with high early mortality reflecting extreme case severity
Nair et al, 2024 ⁵⁵	2,754 CS (216 AS-CS)	AS-CS	Comparative observational	BAV vs TAVR/AVR	BAV may serve as bridge; outcomes poor without definitive valve therapy
Debry et al, 2018 ⁵⁶	44	AS-CS	2-center cohort	BAV timing (≤48 h vs later)	Earlier BAV associated with better outcomes; overall mortality remains high

Footnotes as in Table 1.
AS = aortic stenosis; BAV = balloon aortic valvuloplasty; MR = mitral regurgitation; TAVR = transcatheter aortic valve replacement; TEER = transcatheter edge-to-edge repair; TVT = transcatheter valve therapy; other abbreviations as in Table 1.

define the optimal timing, lesion selection, and sequencing of staged multivessel intervention in the modern MCS era (Table 1).^{4,12,40-44} Residual knowledge gaps include standardized phenotyping of shock severity and trajectory, and prospective trials or prespecified analyses evaluating staged revascularization strategies integrated with contemporary shock and MCS pathways. For patients with AMI and persistent CS, incomplete revascularization including complex multivessel disease or chronic total occlusions may perpetuate ischemia and hinder myocardial recovery. Prospective trials have failed to confirm myocardial viability and the treatment effect of revascularization; however, in select cases, high-risk percutaneous coronary intervention or staged

chronic total occlusion revascularization can be considered, particularly when there is viable myocardium and ischemia in territories of severely diseased coronary arteries.^{45,46} These decisions must balance procedural risk with potential benefit and are best made by a multidisciplinary heart team.^{12,47} Importantly, the feasibility of viability testing is often limited in the acute setting, necessitating individualized decision making informed by anatomic complexity, hemodynamic reserve, residual ischemic burden, and trajectory on support.⁴⁸

Structural interventions. Structural interventions in CS are supported largely by observational registries and administrative claims data (Table 2).⁴⁹⁻⁵⁶ Registry data suggest that concomitant severe valvular

TABLE 3 Electrophysiology Interventions in CS

Study	n	Population	Study Type	Intervention	Key Finding
VT ablation (often MCS-assisted)					
Ballout et al, 2020 ⁶⁰	21	Shock requiring MCS	Series/cohort	MCS-assisted VT ablation	High acute arrhythmia control; many weaned from MCS; mortality remains significant
Baratto et al, 2016 ⁶¹	64	VT storm ± shock	Cohort	ECMO-supported ablation	ECMO facilitates ablation in unstable VT; favorable intermediate survival in series
CRT in inotrope-dependent HF (adjacent evidence; not shock-specific)					
Al-Shakarchi et al, 2024 ⁶²	386	Inotrope-dependent HF	Systematic review/meta-analysis	CRT/CRT-D	High pooled inotrope-weaning and discharge survival; ~70% 1-y survival reported
Bhattacharya et al, 2010 ⁶³	759 (50 inotrope)	CRT-D recipients	Cohort	CRT-D	Inotrope dependence predicts worse prognosis (context for selection, not efficacy)

Footnotes as in Table 1.
CRT = cardiac resynchronization therapy; CRT-D = cardiac resynchronization therapy with defibrillator; HF = heart failure; VT = ventricular tachycardia; other abbreviations as in Tables 1 and 2.

TABLE 4 Interventional Strategies in CS: Integrative Synthesis of Evidence, Limitations, and Remaining Gaps

Domain/Clinical Question	Highest Available Evidence	Conclusions	Key Limitations	Remaining Gaps
AMI-cardiogenic shock revascularization strategy	SHOCK; CULPRIT-SHOCK	Early revascularization improves outcomes; culprit-only PCI favored initially	Unresolved timing of staged PCI; heterogeneity in MCS use and shock severity	Optimal staged timing and lesion selection in contemporary MCS era
Severe MR with cardiogenic shock: role of TEER	TVT Registry; Medicare/claims data	Feasible in selected patients with signal of benefit	Selection and survivorship bias; heterogeneous timing and selection	Prospective studies defining optimal selection, timing, and outcomes
Severe AS with cardiogenic shock: TAVR vs BAV	TVT Registry; case series	Rescue strategies feasible; mortality remains high	Confounding by indication; phenotype heterogeneity	Comparative evaluation of bridge-to-TAVR vs direct strategies
VT-driven shock: role of ablation	Small series	Feasible with MCS support; arrhythmia control achievable	Very small samples; referral and selection bias	Prospective outcomes and durability data
CRT in inotrope-dependent shock	Observational studies; meta-analyses	Selected patients may be weaned from inotropes	Not shock-specific populations; substantial selection bias	Shock-specific evidence and standardized selection criteria

Footnotes as in [Table 1](#).
Abbreviations as in [Tables 1 and 2](#).

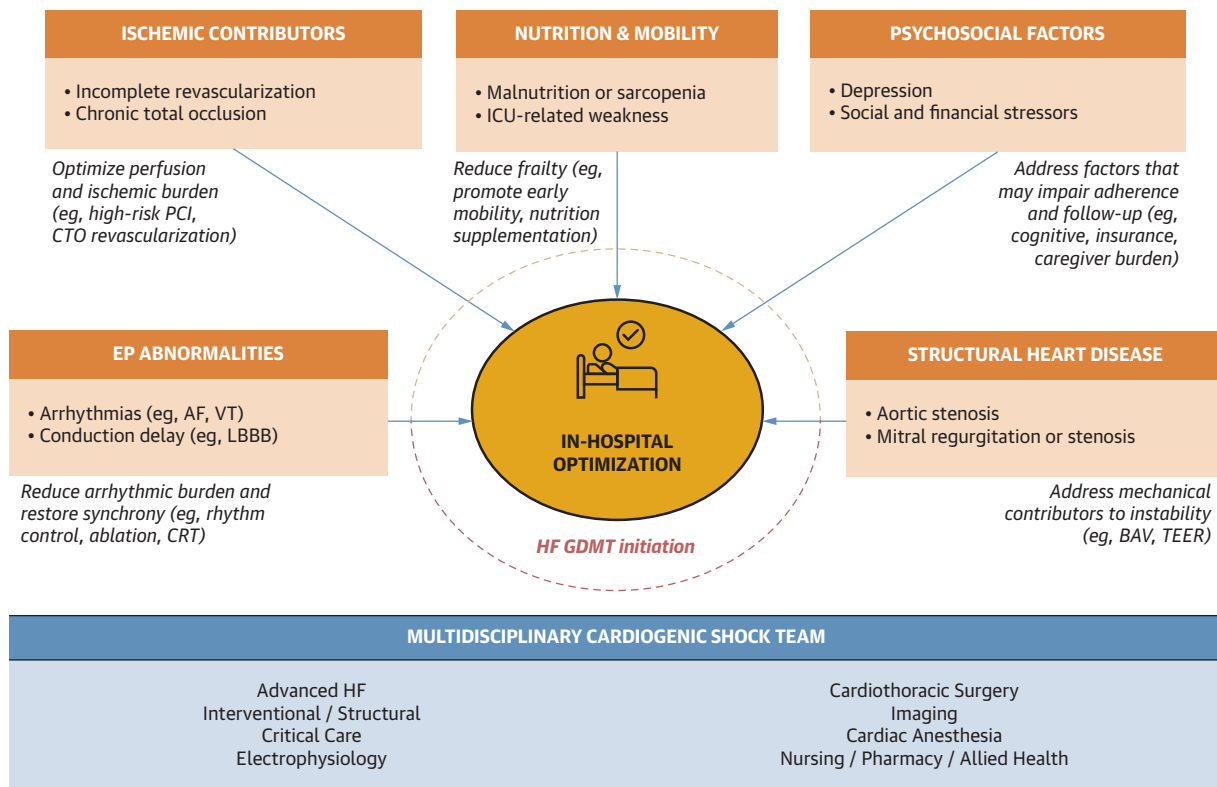
heart disease is associated with worse outcomes in CS and that valve intervention is feasible and associated with acceptable outcomes in select patients.⁵⁷ In patients with critical aortic stenosis, balloon aortic valvuloplasty (BAV) may potentially serve as a bridge to surgical or transcatheter valve replacement in selected hemodynamically unstable patients. Similarly, transcatheter edge-to-edge mitral valve repair has demonstrated feasibility and favorable outcomes in patients with severe mitral regurgitation and CS who are not surgery candidates.^{49,58} In select patients with severe mitral stenosis and favorable anatomy, percutaneous balloon valvuloplasty may also provide hemodynamic benefit.⁵⁹ These interventions, when deployed appropriately, may improve cardiovascular hemodynamics and create a more favorable substrate for HF GDMT initiation to promote myocardial recovery.

Electrophysiologic interventions. Sustained atrial or ventricular arrhythmias are common in CS and may exacerbate hemodynamic instability or delay recovery. Electrophysiology-focused interventions as rescue therapy for CS are informed primarily by small observational series ([Table 3](#)).⁶⁰⁻⁶³ VT ablation with MCS appears feasible for refractory ventricular arrhythmias in selected patients, but available evidence is limited by small sample size, selection bias, and uncertain durability of benefit.⁶⁴ Registry data further suggest that MCS for CS during ablation of atrial arrhythmias (including nonselective atrioventricular node ablation) is feasible and may lead to acceptable outcomes.⁶⁵ For patients not in sinus rhythm, restoration of normal sinus rhythm should be considered if underlying arrhythmia may be contributing to hemodynamic instability. For

those with persistent conduction abnormalities and reduced ejection fraction, early consideration of cardiac resynchronization therapy (CRT) or conduction system pacing may accelerate functional improvement in those who meet eligibility criteria.⁶⁶ In addition, for patients with a high arrhythmic burden, ablation strategies and implantable cardioverter-defibrillator (ICD) planning should be integrated into longitudinal care planning.

Collectively, these interventional strategies, implemented alongside volume optimization, afterload management, and individualized HF GDMT initiation, underscore the need for an etiology-specific, multidisciplinary, and physiology-driven approach. This period should not be conceptualized merely as a bridge to discharge; rather, it is a therapeutic interval in which targeted intervention may meaningfully alter long-term trajectory. At the same time, it is essential to acknowledge that much of the evidence supporting these strategies in CS survivors remains observational and is influenced by selection and survivorship bias, reinforcing the need for standardized eligibility criteria, procedural timing frameworks, and prospective evaluation.

Need for protocolized in-patient frameworks and structured transitions. One of the most pressing needs is the development and implementation of protocolized in-hospital frameworks that outline best practices for diagnostic reassessment, therapeutic optimization, tMCS weaning, and discharge planning. Emerging observational data in selected survivors of tMCS-supported CS suggest that early in-hospital initiation of HF GDMT before ICU discharge may be feasible and associated with improved postdischarge implementation and

FIGURE 2 Targeted Interventions During the In-Hospital Postrescue Phase of Cardiogenic Shock

The in-hospital postrescue phase provides an opportunity to address optimization of care, clinical trajectory, and reversible contributors to myocardial dysfunction before hospital discharge. Key domains include: 1) ischemic contributors, such as incomplete revascularization or chronic total occlusions (CTOs), where targeted revascularization strategies (eg, high-risk percutaneous coronary intervention [PCI], CTO revascularization) can optimize myocardial perfusion and reduce ischemic burden; 2) electrophysiologic (EP) abnormalities, including atrial fibrillation (AF), ventricular tachycardia (VT), and conduction delays (eg, left bundle branch block [LBBB]), where rhythm control, ablation, or cardiac resynchronization therapy (CRT) may restore synchrony and reduce arrhythmic burden; 3) structural heart disease, such as aortic stenosis or significant mitral valve disease, where interventions (eg, balloon aortic valvuloplasty [BAV], transcatheter edge-to-edge repair [TEER]) can address mechanical contributors to instability; 4) nutrition and mobility, including early mobilization, nutritional optimization to support recovery, and prevention of muscle wasting and frailty, which are critical for restoring functional capacity; and 5) psychosocial factors, including depression, post-traumatic stress, cognitive impairment, financial strain, lack of insurance, and caregiver burden, which may impair adherence, follow-up, and quality of life if left unaddressed. Proactive screening and early psychosocial intervention should be integrated into remission-phase protocols to improve long-term outcomes. Optimal management requires coordination within a multidisciplinary heart team including advanced heart failure, interventional and structural cardiology, critical care, electrophysiology, cardiothoracic surgery, imaging, cardiac anesthesia, nursing, pharmacy, and allied health professionals. AF = atrial fibrillation; GDMT = guideline-directed medical therapy; HF = heart failure; ICU = intensive care unit; TEER = transcatheter edge-to-edge repair; VT = ventricular tachycardia.

6-month outcomes, supporting the need for physiology-guided protocols and prospective validation.⁶⁷ However, specific protocols for drug titration, sequencing, and timing relative to inotrope or device discontinuation remain limited, with decisions largely left to clinician discretion.^{9,68,69}

In the absence of definitive evidence supporting a uniform sequencing strategy in post-CS patients, HF GDMT initiation during in-hospital stabilization should be treated as a practical bedside roadmap rather than a prescriptive protocol. One pragmatic sequencing strategy is to: 1) achieve hemodynamic

stability with declining vasoactive needs and near-normal perfusion markers; 2) initiate therapies with minimal blood pressure effect (eg, sodium-glucose cotransporter 2 inhibitors, then mineralocorticoid receptor antagonists when potassium and renal function permit); 3) introduce low-dose angiotensin-converting enzyme inhibitor, angiotensin receptor blocker, or angiotensin receptor-neprilysin inhibitor only when hypotension risk is low; and 4) defer beta-blockers until inotropes are discontinued (with selected exceptions on milrinone). This approach prioritizes safety, minimizes weaning

failure, and acknowledges that HF GDMT tolerance in CS is driven more by physiology than by strict timelines.⁷⁰ Importantly, however, prospective trials validating these strategies are lacking, and no randomized studies have evaluated different initiation or intensification approaches tailored to the post-CS population. This knowledge gap is particularly important given the heterogeneity of patients recovering from shock, ranging from those with preexisting HF on chronic therapy to de novo presentations with uncertain tolerance and response to HF GDMT. The INTEGRATE (Initiation and Titration of Guideline-Directed Medical Therapy in Heart Failure Cardiogenic Shock With Impella 5.5 for Cardiac Recovery; [NCT06965504](#)) trial, should help to clarify this important and unresolved area.

An additional and increasingly common barrier to in-patient CS care is premature hospital discharge without a structured transition-of-care framework, driven in part by administrative pressures, capacity constraints, and reimbursement incentives. In practice, this often truncates the period when key survivorship-defining processes should occur. Complete decongestion, initiation and optimization of HF GDMT, multidisciplinary reassessment of reversible drivers, trajectory assessment, and alignment on destination therapy planning should all be accomplished before discharge. Standardized post-CS discharge bundles and shock-specific transition pathways remain uncommon,⁶ and postdischarge disease management infrastructure is rarely calibrated to the multisystem consequences of critical illness. Importantly, structured transitions must also account for noncardiac determinants of survivorship, which frequently emerge during hospitalization and directly influence adherence, follow-up reliability, and functional reintegration.^{6,7} Depression, post-traumatic stress symptoms, cognitive impairment, and social and financial stressors (eg, caregiver burden, insurance disruption, loss of income) are prevalent yet underrecognized, and most centers lack systematic workflows for screening and early intervention before discharge. To bridge these gaps, partnerships across inpatient disciplines, post-acute care facilities, skilled nursing programs, and telemonitoring networks that can support surveillance for early signs of deterioration are urgently needed.

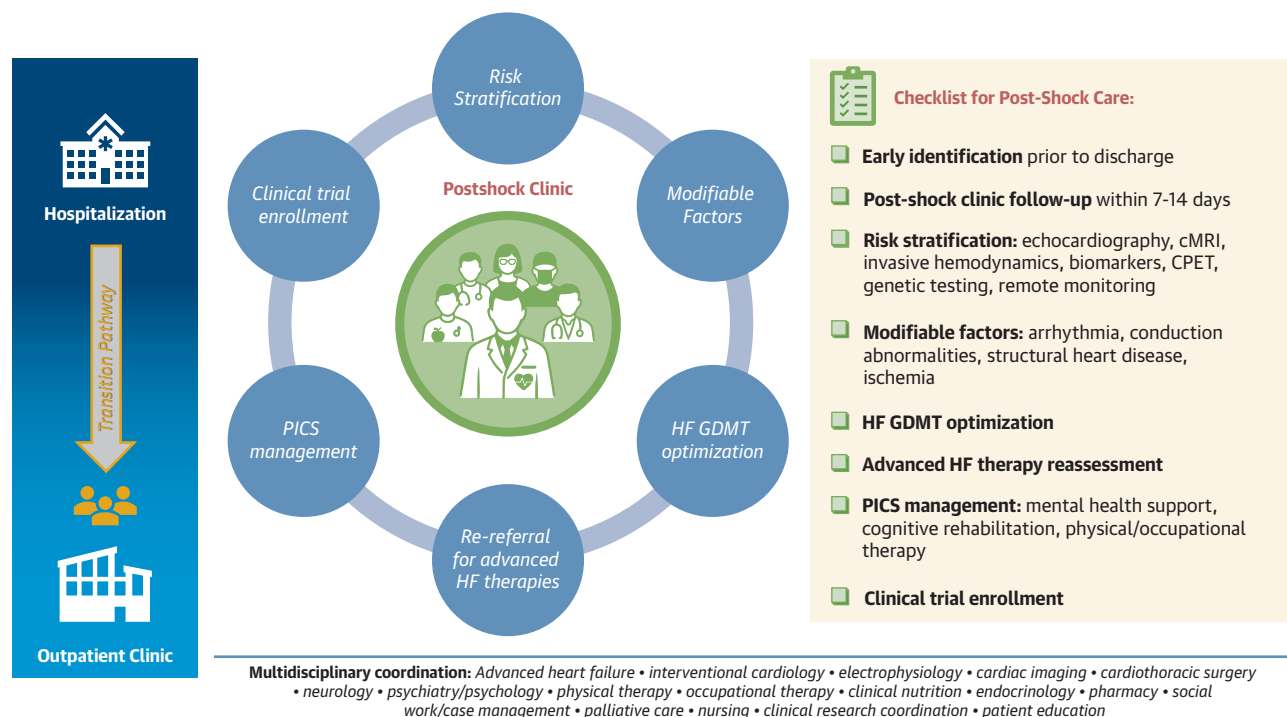
POSTDISCHARGE SHOCK CARE. Long-term trajectories after CS are inherently heterogeneous. Some patients, particularly those with potentially reversible etiologies, including revascularized AMI, acute myocarditis, tachycardia-mediated, peripartum, or stress-induced cardiomyopathy, may experience

near-normalization of ventricular function and restoration of functional independence. Others, such as those with underlying chronic HF or non-revascularized ischemic cardiomyopathy, may enter a trajectory of residual ventricular dysfunction or progressive functional impairment despite early stabilization.^{17,71,72}

From a postdischarge survivorship standpoint, patients who achieve native heart survival often follow 1 of 3 pathways: 1) early stabilization with complete myocardial and functional restitution (recovery); 2) persistent ventricular dysfunction with clinical stability but ongoing vulnerability (remission phenotype); or 3) recurrent decompensation and hemodynamic limitation requiring accelerated evaluation for dMCS, heart or multiorgan transplant, or palliative transitions. Regardless of trajectory, nearly all CS survivors would benefit from comprehensive postshock care to reduce rehospitalizations, address noncardiac sequelae, and preserve quality of life ([Figure 3](#)).

Risk stratification and modifiable determinants of recovery. A central element of long-term postshock care is systematic cardiovascular reassessment coupled with dynamic longitudinal risk stratification and targeted modification of drivers of recurrent instability. Risk assessment after CS should be conceptualized as an iterative process rather than a single postdischarge evaluation, with the goal of detecting latent physiologic vulnerability (persistent congestion, low-output physiology, arrhythmia risk, and end-organ limitations) early enough to intervene. The development of shock-specific outpatient recovery clinics, analogously to transplant and LVAD programs, may provide an ideal platform for structured postdischarge care, especially when co-located within centers with advanced HF expertise. As part of postdischarge risk stratification, cardiac and noncardiac domains should be assessed as multidimensional determinants of long-term stability. By applying a structured framework that integrates imaging, biomarkers, HF risk tools, genetics, residual structural and electrophysiologic assessment, and functional, neurocognitive, and psychosocial domains, clinicians can better determine prognosis, personalize care pathways, and identify inflection points that warrant escalation to advanced HF therapies.

1. Imaging. Transthoracic echocardiography remains essential for serial monitoring of biventricular systolic function, residual valvular disease, estimation of filling pressures, pulmonary pressures, and longitudinal remodeling. Cardiac magnetic

FIGURE 3 Structured Postdischarge Care Model for Cardiogenic Shock Survivors

Postdischarge care involves hospital-based identification with coordinated transition to outpatient follow-up. The centralized postshock clinic integrates risk stratification (echocardiography, cardiac magnetic resonance imaging [cMRI], biomarkers, invasive hemodynamics, cardiopulmonary exercise testing [CPET], cognitive and frailty assessment), management of modifiable factors (structural heart disease, arrhythmias, metabolic and psychosocial determinants), heart failure (HF) guideline-directed medical therapy (GDMT) optimization, advanced HF therapy candidacy reassessment (left ventricular assist device, heart transplant), post-intensive care syndrome (PICS) management, and clinical trial enrollment. Multidisciplinary coordination includes advanced HF, interventional cardiology, electrophysiology, cardiac imaging, cardiothoracic surgery, neurology, psychiatry/psychology, neuropsychology, physical medicine and rehabilitation, physical and occupational therapy, nutrition, endocrinology, pharmacy, social work, palliative care, and clinical research.

resonance imaging (cMRI) provides incremental mechanistic and prognostic insight through tissue characterization, quantification of fibrosis, and late gadolinium enhancement, which may help to distinguish potentially reversible myocardial injury from fixed scar and refine arrhythmic risk assessment. However, cMRI feasibility in the peri-shock period is often limited by hemodynamic instability and concomitant MCS, and interpretation may be confounded by myocardial edema, evolving necrosis, or microvascular obstruction. Furthermore, although cMRI is often used to assess viability in ischemic cardiomyopathy, evidence that viability testing reliably identifies patients who derive a survival benefit from revascularization is limited.^{48,73} Accordingly, cMRI should be positioned as a tool for postshock risk stratification and mechanistic inference rather than a definitive determinant of acute therapeutic decisions.

- Biomarkers.** Serial N-terminal pro-B-type natriuretic peptide and high-sensitivity troponin can reflect ongoing myocardial stress and subclinical decompensation.^{23,74} Dynamic changes over time are often more informative than single measurements and can identify patients at risk for deterioration despite apparent clinical stability. Emerging biomarkers, including circulating dipeptidyl peptidase 3, adrenomedullin, soluble ST2, and growth differentiation factor 15, have shown prognostic utility in small cohorts and could augment traditional risk models.¹¹
- Cardiopulmonary exercise testing and hemodynamics.** Cardiopulmonary exercise testing offers objective metrics of functional capacity and correlates strongly with prognosis.⁷⁵ Invasive hemodynamic reassessment (right heart catheterization), particularly in patients with persistent symptoms or borderline clinical status, enables precise measurement of cardiac index, pulmonary

TABLE 5 Core Elements for Establishing a Postshock Care Program

Domain	Key Components	Operational Strategy
Multidisciplinary team	Advanced HF, critical care, cardiology (interventional, structural, EP), rehab, mental health, pharmacy	Define team roles; schedule joint case conferences; embed within HF program
Clinic infrastructure	Dedicated postshock clinic or embedded model within an advanced HF clinic	Establish routine follow-up intervals; allocate nursing/pharmacy support for GDMT titration
HF GDMT optimization	RAASI, beta-blocker, MRA, SGLT2i	Protocol-driven up-titration; pharmacist-led education; track medication tolerance and barriers
Risk stratification	Echo, cMRI, biomarkers (NT-proBNP, hs-Tn), CPET, RHC, genetic testing	Create standardized protocol for serial assessment at 30, 90, and 180 days after discharge
Arrhythmia management	Holter/loop recorder, CRT/ICD consideration, EP consultation	Initiate rhythm surveillance; define criteria for EP referral and device evaluation
Structural reassessment	Valvular disease surveillance (TTE/TEE), revascularization assessment	Use imaging follow-up and viability studies to guide staged intervention or reassess candidacy
Functional/cognitive assessment	Frailty, PICS screening, MoCA/DSST, GAD-7, PHQ-9	Implement during clinic visits; integrate referrals to PM&R, neurology, psychology
Rehabilitation services	Physical/occupational therapy, nutritional counseling	Develop referral pathways and individualized rehab plans based on deficits and patient goals
Patient and caregiver engagement	Education, psychosocial support, medication literacy	Provide discharge packets; integrate social work and patient navigator roles
Remote monitoring and technology	PA pressure monitoring, digital apps, virtual visits	Identify candidates for remote monitoring; partner with remote HF monitoring services
Data collection and research	Longitudinal registry, patient-reported outcomes	Collect standardized data; consider registry-embedded trials; define QI metrics

cMRI = cardiac magnetic resonance imaging; CPET = cardiopulmonary exercise testing; CRT = cardiac resynchronization therapy; DSST = Digit Symbol Substitution Test; EP = electrophysiology; GAD-7 = Generalized Anxiety Disorder 7-item Scale; GDMT = guideline-directed medical therapy; HF = heart failure; hs-Tn = high-sensitivity troponin; ICD = implantable cardioverter-defibrillator; MoCA = Montreal Cognitive Assessment; MRA = mineralocorticoid receptor antagonist; NT-proBNP = N-terminal pro-B-type natriuretic peptide; PA = pulmonary artery; PHQ-9 = Patient Health Questionnaire-9; PICS = post-intensive care syndrome; PM&R = physical medicine and rehabilitation; QI = quality improvement; RAASI = renin-angiotensin-aldosterone system inhibitor; RHC = right heart catheterization; SGLT2i = sodium-glucose cotransporter 2 inhibitor; TEE = transesophageal echocardiography; TTE = transthoracic echocardiography.

pressures, and right-side function, which are key variables in advanced HF decision making.

- Genetic testing.** In nonischemic cardiomyopathies, genetic testing remains underutilized yet highly informative. Truncating variants in *TTN* are the most frequently observed, while *LMNA*, *RBM20*, *DSP*, and *FLNC* also are frequently implicated in familial or high risk phenotypes. These variants are associated with increased arrhythmogenic risk and worse prognosis, and may guide tailored surveillance.^{76,77}
- Residual structural and electrophysiologic assessment.** Residual structural and electrophysiologic abnormalities are common after CS. Mitral or

aortic valve disease, either newly identified or exacerbated during the shock episode, may warrant intervention once patients are stabilized. The therapeutic spectrum includes percutaneous approaches (eg, transcatheter edge-to-edge repair, transcatheter aortic valve replacement) and surgical strategies (eg, valve surgery, coronary artery bypass grafting), with the optimal approach determined by a multidisciplinary heart team integrating patient comorbidity, procedural risk, and anticipated benefit.⁷⁸ Similarly, arrhythmias encountered during the acute course necessitate postdischarge rhythm monitoring with ambulatory telemetry or implantable loop recorders.^{24,60,79} These data can guide individualized decisions regarding antiarrhythmic therapy, catheter ablation, and device implantation (eg, ICD, CRT), with the overarching goal of reducing recurrence, optimizing functional status, and, in selected cases, facilitating ventricular recovery.

- Functional, neurocognitive, and psychosocial determinants.** As previously mentioned, PICS, which includes physical weakness, cognitive dysfunction, and psychologic distress, is prevalent among CS survivors and significantly impairs reintegration and long-term quality of life.^{6,7} Validated screening tools, such as the Mini-Montreal Cognitive Assessment (MoCA), Digit Symbol Substitution Test (DSST), Generalized Anxiety Disorder 7-Item Scale (GAD-7), and Patient Health Questionnaire-9 (PHQ-9), should be used to detect cognitive impairment, depression, anxiety, and post-traumatic stress disorder. Frailty assessments, including gait speed, grip strength, and Fried criteria, can identify barriers to rehabilitation or advanced therapy candidacy. Nutritional status, sarcopenia, and social determinants of health (eg, housing stability, caregiver support) also inform both prognosis and recovery potential.

Multidisciplinary postshock clinics: a promising paradigm shift in CS care. The postshock experience is currently often characterized by fragmented care, therapeutic inertia, and high risk for clinical deterioration. Robust care pathways exist for patients discharged after LVAD implantation and transplantation, but similar longitudinal models for CS survivors without heart replacement therapies (LVAD or transplantation) remain uncommon. In this context, dedicated multidisciplinary postshock care clinics have emerged as a promising paradigm to optimize long-term outcomes (Table 5). Their value lies not only in clinical consolidation, but also in operationalizing survivorship through standardized

workflows, phase-specific targets, and longitudinal trajectory assessment.

Despite their promise and potential, the implementation of post-CS care clinics remains limited, and there are no formal guidelines recommending their creation or outlining their structure. There is also a lack of dedicated reimbursement models for multidisciplinary longitudinal care, barriers that must be addressed to expand this paradigm nationally. To address this gap, we propose a pragmatic multidisciplinary framework for postshock clinics designed to standardize survivorship care, support structured risk reassessment and intervention, and improve outcomes among CS survivors (Figure 3).

At their core, postshock clinics should operationalize 4 interdependent functions. First, they should reduce therapeutic inertia by enabling accelerated HF GDMT implementation and titration according to structured protocols, dedicated staffing, and multidisciplinary care engagement.⁸⁰ Nurse-led titration programs and pharmacist-supported medication reconciliation can facilitate safe initiation and up-titration of renin-angiotensin-aldosterone system inhibition and angiotensin receptor-neprilysin inhibitor strategies, beta-blockers, mineralocorticoid receptor antagonists, and sodium-glucose cotransporter 2 inhibitors, alongside individualized diuretic adjustments and frequent laboratory monitoring. Importantly, this early postdischarge interval is also the optimal window to address medication access barriers, cost-related nonadherence, and patient education gaps that meaningfully influence downstream outcomes.

Second, postshock clinics should function as hubs for multidisciplinary cardiovascular reassessment and intervention.⁸¹ Integration of advanced HF, electrophysiology, structural heart, and interventional cardiology expertise enables systematic evaluation for staged revascularization, rhythm control strategies, device therapy (ICD, CRT), and structural intervention when physiologically appropriate. Importantly, this is increasingly extendable through remote monitoring and digital health infrastructure, enabling scalable proactive surveillance rather than reactive hospitalization-based care.⁸² Implantable pulmonary artery pressure monitoring and rhythm surveillance systems (eg, implantable loop recorders) can detect early physiologic deterioration, guide individualized diuretic and HF GDMT optimization strategies, and support earlier escalation when warranted. In parallel, wearable-derived physiologic and functional signals (eg, resting heart rate, activity burden, step counts, recovery metrics) represent an emerging opportunity to longitudinally quantify

health status trajectories and integrate patient-centered outcomes into survivorship care. This integrated multidisciplinary digital platform also supports early reassessment of candidacy for LVAD or transplantation, particularly among patients initially limited by frailty, psychosocial instability, or reversible end-organ dysfunction, barriers that may improve with structured survivorship interventions.

Third, postshock clinics should embed PICS-focused survivorship care into cardiovascular follow-up. Systematic screening for cognitive impairment, depression, anxiety, and post-traumatic stress symptoms, paired with structured referral pathways to physical medicine and rehabilitation, physical therapy, occupational therapy, neuropsychology, and social work, transforms survivorship from an implicit expectation to an explicit care target.^{6,81} This infrastructure is particularly important for patients supported with VA-ECMO, prolonged mechanical ventilation, or extended ICU stays, in whom neuromuscular weakness, delirium-related cognitive impairment, and prolonged deconditioning may be profound and directly linked to rehospitalization risk.

Finally, the postshock clinic can identify candidates for ongoing clinical trials which will provide important insight into postshock outcomes.¹ Several contemporary studies are beginning to systematically evaluate postdischarge outcomes after CS. For example, HALO-Shock (Hemodynamic Monitoring to Prevent Adverse Events foLLowing cardiOgenic Shock Trial; [NCT04419480](#)) is randomizing patients stabilized after CS to implantable pulmonary artery pressure monitoring, with a hierarchic 6-month primary endpoint that incorporates both clinical events and patient-reported outcomes (PROs), including HF-related QoL. The Cardiogenic Shock Working Group (CSWG) has initiated efforts to assess longitudinal postdischarge HRQoL in multicenter cohorts, providing real-world insights into survivorship trajectories. In parallel, VANQUISH-SHOCK (Multi-center Collaborative to Enhance Quality and Outcomes in the Management of Cardiogenic Shock; [NCT05185492](#)) is prospectively evaluating health status, QoL, and functional disability at 6 and 12 months after discharge, aiming to capture both the clinical and the patient-centered impact of shock beyond the index hospitalization. Collectively, these initiatives reflect a paradigm shift from focusing solely on in-hospital survival toward recognizing QoL, functional capacity, and long-term disability as equally critical endpoints in CS research.

Importantly, postshock clinics represent a test bed for research and innovation. Because post-CS care

TABLE 6 Strategic Priorities to Advance Cardiogenic Shock (CS) Survivorship Research and Care

Strategic Domain	Key Initiatives	Scientific Rationale and Impact
1. Standardize postshock care	- Develop dedicated post-CS registries - Conduct pragmatic, registry-embedded RCTs - Define core datasets and longitudinal follow-up intervals	Establishes structured infrastructure to evaluate recovery trajectories, compare care models, and identify predictors of long-term outcomes
2. Expand clinical trial endpoints	- Integrate validated PROs and HRQoL tools (eg, KCCQ, EQ-5D, MoCA, PHQ-9) - Develop composite endpoints reflecting survival and functional recovery	Aligns clinical research with patient-centered goals; captures functional and cognitive outcomes beyond traditional mortality-based endpoints
3. Define long-term trajectories	- Build longitudinal cohorts with serial imaging, hemodynamics, biomarkers - Link with EHRs, claims, and remote data (eg, wearables)	Enables recovery phenotyping, facilitates early risk stratification, and informs intensity and timing of follow-up and intervention
4. Apply precision medicine approaches	- Apply multi-omics (proteomics, metabolomics, transcriptomics) - Deploy AI for dynamic risk modeling	Identifies mechanistic subtypes, uncovers novel therapeutic targets, and enables individualized care pathways across the shock recovery continuum
5. Foster multicenter collaboration and networks	- Strengthen global consortia (eg, CSWG, CCTN, AHA, FRENDSHOCK) - Build hub-and-spoke systems for trial access - Engage patient stakeholders in research design	Promotes data harmonization, increases generalizability, and accelerates translation through shared infrastructure and patient-informed research priorities

AHA = American Heart Association; AI = artificial intelligence; CCTN = Critical Care Cardiology Trials Network; cMRI = cardiac magnetic resonance imaging; CSWG = Cardiogenic Shock Working Group; EHR = electronic health record; EQ-5D = EuroQol 5-Dimensions; FRENDSHOCK = French Cardiogenic Shock Registry; HRQoL = health-related quality of life; KCCQ = Kansas City Cardiomyopathy Questionnaire; LGE = late gadolinium enhancement; MoCA = Montreal Cognitive Assessment; PHQ-9 = Patient Health Questionnaire-9; PRO = patient-reported outcome; RCT = randomized controlled trial.

remains an underexplored domain, postshock clinics can serve as prospective registries for recovery phenotypes, therapeutic response, and long-term outcomes. Such data will be essential to defining best practices, establishing quality benchmarks, and informing future guidelines.

From theory to practice: how to operationalize postshock clinics? To operationalize this model, institutions should implement a standardized tiered postshock clinic infrastructure, analogously to oncology survivorship programs and post-ICU recovery clinics,^{6,81,83,84} with: 1) defined referral triggers; 2) prespecified visit cadence; 3) multidisciplinary staffing; and 4) measurable quality benchmarks.

Pragmatically, all patients discharged after CS without dmCS, transplantation, or hospice should be automatically referred at discharge, with “high-intensity” follow-up prioritized for those with persistent systolic dysfunction, ongoing congestion,

recurrent arrhythmias, residual end-organ dysfunction, frailty, or prolonged ICU or MCS exposure.

A pragmatic approach is to define structured postdischarge reassessment checkpoints: early (7-14 days), intermediate (4-6 weeks), and consolidation (3-6 months). Early reassessment prioritizes congestion status, perfusion adequacy, medication tolerance, rhythm surveillance, and psychosocial screening. Intermediate reassessment focuses on imaging- and biomarker-informed trajectory classification and optimization. Consolidation reassessment integrates functional capacity and, when indicated, invasive hemodynamic reassessment and formal advanced HF therapy candidacy reassessment.

Core staffing should include an advanced HF provider, nurse navigator, HF pharmacist, and social work embedded into clinic operations, with protocolized access to electrophysiology, structural/interventional cardiology, physical medicine and rehabilitation, rehabilitation, and neuropsychology (Mini-MoCA, DSST, GAD-7, and PHQ-9) through predefined referral pathways.

Finally, to enable scalability and benchmarking, postshock clinics should adopt standardized documentation templates and track program-level metrics including early (≤ 30 -day) rehospitalization, HF GDMT implementation rates and dose intensity, volume status at follow-up, arrhythmia burden, functional recovery trajectories, patient-reported outcomes (eg, Kansas City Cardiomyopathy Questionnaire [KCCQ] and EuroQol-5D), and rates and timing of escalation to durable LVAD or transplantation. Embedding these clinics within prospective registries and learning health system frameworks will accelerate refinement of survivorship care, enable cross-institutional benchmarking, and support pragmatic trials to define evidence-based postshock care standards.

FUTURE DIRECTIONS IN CS SURVIVORSHIP AND LONGITUDINAL CARE

The evolving paradigm of CS care necessitates a reframing of success, one that extends beyond short-term survival to embrace longitudinal recovery, functional reintegration, and quality of life. To achieve this vision, future investigation must address 5 interrelated domains (Table 6): 1) standardizing postshock care through dedicated registries and pragmatic clinical trials; 2) expanding clinical trial endpoints to include HRQoL and PROs; 3) defining long-term trajectories using real-world and registry-based data; 4) applying precision medicine tools including biomarkers, “shock-omics,” imaging,

and artificial intelligence (AI)-driven models; and 5) fostering multicenter collaboration through global research consortia.

STANDARDIZING POSTSHOCK CARE THROUGH DEDICATED REGISTRIES AND CLINICAL TRIALS.

Future efforts should prioritize the development of dedicated postshock registries that include standardized case definitions, core data elements, and prespecified follow-up intervals extending to at least 1 year. These registries can serve dual roles as data repositories as well as platforms for pragmatic, registry-embedded randomized controlled trials (RCTs). Registry-based RCTs have successfully improved efficiency and external validity in other cardiovascular domains and offer a cost-effective means to evaluate novel care models, technologies, or therapeutic strategies in real-world settings.^{85,86} The integration of electronic health record (EHR) linkages to capture outpatient visits, rehospitalizations, and mortality will be essential to enhance data completeness. Adoption of harmonized core outcome sets, modeled on frameworks like COMET (Context, Objectives, Materials, Execution, and Team Dynamics) and guided by recent Shock Academic Research Consortium and Percutaneous Ventricular Assist Device Academic Research Consortium initiatives, will enable benchmarking of outcomes and comparison across institutions and populations.⁸⁷

EXPANDING CLINICAL TRIAL ENDPOINTS TO INCLUDE HRQoL AND PROs.

Traditional CS trials fail to capture the multidimensional burden faced by survivors, including PICS, cognitive decline, emotional distress, and functional disability. Incorporating validated HRQoL and PRO instruments, such as the KCCQ, EuroQoL-5D, MoCA, and PHQ-9, will enable a more holistic assessment of recovery.^{6,88,89} Future clinical trials and registries should include composite endpoints that integrate both survival and QoL domains, thus aligning trial outcomes with patient-centered priorities. A core comprehensive outcome set will be needed to standardize benchmarks to promote both sustainability and scalability of postshock clinics.

DEFINING LONG-TERM TRAJECTORIES WITH POSTSHOCK REGISTRIES AND REAL-WORLD DATA.

Establishing longitudinal cohorts with serial clinical, biomarker, hemodynamic, and imaging data is essential to map recovery trajectories and identify phenotypes with different risks of relapse or progression. Real-world data sources, including EHRs, administrative claims, and wearable devices, can complement prospective registries by offering granular insight into outpatient recovery, physical

activity, and readmissions. Linking CS events to existing HF registries may also clarify the downstream impact of transplant, LVAD, and optimal medical therapy utilization. These data can inform the timing and intensity of follow-up, rehabilitation, and screening protocols.

APPLYING PRECISION MEDICINE: “SHOCK-OMICS”

AND AI. Precision medicine offers transformative potential for individualized care pathways in CS.³⁰ Multidimensional “shock-omics” approaches integrating transcriptomic, proteomic, and metabolomic signatures may uncover novel phenotypes or recovery pathways.⁹⁰ AI and machine learning models applied to multimodal datasets, clinical, hemodynamic, imaging, and biochemical, may provide dynamic risk prediction across the continuum of care, including early recognition of CS patients. For example, postdischarge risk calculators could stratify patients for high-touch follow-up vs standard care, and in-hospital algorithms could recalibrate mortality risk and recovery potential in real time. The integration of these technologies into future trials and registries should be accompanied by rigorous validation and transparency to ensure equitable application across patient populations.

FOSTERING MULTICENTER COLLABORATION AND GLOBAL RESEARCH CONSORTIA.

Multicenter collaboration is indispensable for accelerating innovation in CS survivorship. Existing networks, such as the Cardiogenic Shock Working Group, Critical Care Cardiology Trials Network, and French Observatory on the Management of Cardiogenic Shock, among others, have demonstrated that global consortia can generate meaningful data and catalyze practice change. The development of regional tier-based networks linking tertiary CS centers with referring hospitals can support prospective registry enrollment, enhance care transitions, and extend clinical trial access to broader populations. Public-private partnerships, particularly those linking device manufacturers with academic networks, will be critical for generating long-term data on device performance, complications, and functional recovery.

Future initiatives must also prioritize the inclusion of patient and caregiver voices in trial design and outcome selection. Stakeholder engagement will ensure alignment with real-world priorities, enhance study feasibility, and strengthen the translational impact of research.

CONCLUSIONS

With improving survival after CS, the absence of structured longitudinal care pathways has become a

critical gap in the continuum of care. This review outlines a proposed framework for post-CS survivorship, emphasizing the need for standardized in-hospital protocols, risk-stratification strategies, and multidisciplinary outpatient models to guide medical therapy, rehabilitation, and psychosocial support. Redefining therapeutic success in CS must extend beyond short-term hemodynamic stabilization to include long-term functional recovery, HRQoL, and reduction in recurrent hospitalizations. Future investigation should prioritize core outcome sets inclusive of PROs, real-world data integration, and precision phenotyping to inform tailored interventions. Advancing survivorship in CS will require coordinated multicenter efforts and the development of dedicated clinical infrastructure to support this emerging paradigm.

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REFERENCES

- Hall EJ, Agarwal S, Cullum CM, Sinha SS, Ely EW, Farr MA. Survivorship after cardiogenic shock. *Circulation*. 2025;151:257-271. <https://doi.org/10.1161/circulationaha.124.068203>
- Hall EJ, Schaffert J, Agarwal S, et al. Cognitive function and patient-reported outcomes after cardiogenic shock. *J Am Coll Cardiol*. 2025;85:1604-1607. <https://doi.org/10.1016/j.jacc.2025.02.023>
- Mehta A, Vavilin I, Nguyen AH, et al. Contemporary approach to cardiogenic shock care: a state of the art review. *Front Cardiovasc Med*. 2024;11:1354158. <https://doi.org/10.3389/fcvm.2024.1354158>
- Abraham J, Blumer V, Burkhoff D, et al. Heart failure-related cardiogenic shock: pathophysiology, evaluation and management considerations. *J Card Fail*. 2021;27:1126-1140. <https://doi.org/10.1016/j.cardfail.2021.08.010>
- Kanwar MK, Billia F, Randhawa V, et al. Heart failure related cardiogenic shock: an ISHLT consensus conference content summary. *J Heart Lung Transplant*. 2024;43:189-203. <https://doi.org/10.1016/j.healun.2023.09.014>
- Thiele H, Akin I, Sandri M, et al. PCI strategies in patients with acute myocardial infarction and cardiogenic shock. *N Engl J Med*. 2017;377:2419-2432. <https://doi.org/10.1056/nejmoa1710261>
- Sinha SS, Rosner CM, Tehrani BN, et al. Cardiogenic shock from heart failure versus acute myocardial infarction: clinical characteristics, hospital course, and 1-year outcomes. *Circ Heart Fail*. 2022;15:e009279. <https://doi.org/10.1161/circheartfailure.121.009279>
- Møller JE, Engstrøm T, Jensen LO, et al. Microaxial flow pump or standard care in infarct-related cardiogenic shock. *N Engl J Med*. 2024;390:1382-1393. <https://doi.org/10.1056/nejmoa2312572>
- Genstler J, Miller R, Chiu ST, Spinelli KJ, Abraham J. Initiating medical therapy in cardiogenic shock patients on temporary mechanical circulatory support. *JACC Heart Fail*. 2025;13:102534. <https://doi.org/10.1016/j.jchf.2025.102534>
- Thiele H, Zeymer U, Neumann F-J, et al. Intraaortic balloon support for myocardial infarction with cardiogenic shock. *N Engl J Med*. 2012;367:1287-1296. <https://doi.org/10.1056/nejmoa1208410>
- Kimmoun A, O'Brien C, Blumer V, et al. Optimising trial design for cardiogenic shock: insights from the sixth Critical Care Clinical Trialists Workshop. *Lancet Respir Med*. 2025;13:833-842. [https://doi.org/10.1016/s2213-2600\(25\)00084-0](https://doi.org/10.1016/s2213-2600(25)00084-0)
- Hochman JS, Sleeper LA, Webb JG, et al, SHOCK Investigators. Early revascularization in acute myocardial infarction complicated by cardiogenic shock. Should we emergently revascularize occluded coronaries for cardiogenic shock? *N Engl J Med*. 1999;341:625-634. <https://doi.org/10.1056/nejm199908263410901>
- Hiser SL, Fatima A, Ali M, Needham DM. Post-intensive care syndrome (PICS): recent updates. *J Intensive Care*. 2023;11:23. <https://doi.org/10.1186/s40560-023-00670-7>
- Blumer V. Recovery in cardiogenic shock: why terminology matters. *J Card Fail*. Published online September 27, 2025. <https://doi.org/10.1016/j.cardfail.2025.09.027>
- Eisenhauer EA, Therasse P, Bogaerts J, et al. New response evaluation criteria in solid tumours: revised RECIST guideline (version 1.1). *Eur J Cancer*. 2009;45:228-247. <https://doi.org/10.1016/j.ejca.2008.10.026>
- John K, Khalif A, Tsukashita M, Kanwar MK. Myocardial recovery in cardiogenic shock. *Methodist Debaque Cardiovasc J*. 2024;20:64-75. <https://doi.org/10.14797/mdcvj.1383>
- Kanwar MK, Selzman CH, Ton VK, et al. Clinical myocardial recovery in advanced heart failure with long term left ventricular assist device support. *J Heart Lung Transplant*. 2022;41:1324-1334. <https://doi.org/10.1016/j.healun.2022.05.015>
- Park AC, Mann DL. The pathobiology of myocardial recovery and remission: from animal models to clinical observations in heart failure patients. *Methodist Debaque Cardiovasc J*. 2024;20:16-30. <https://doi.org/10.14797/mdcvj.1389>
- Mann DL, Barger PM, Burkhoff D. Myocardial recovery and the failing heart: myth, magic, or molecular target? *J Am Coll Cardiol*. 2012;60:2465-2472. <https://doi.org/10.1016/j.jacc.2012.06.062>
- Mann DL, Burkhoff D. Is myocardial recovery possible and how do you measure it? *Curr Cardiol Rep*. 2012;14:293-298. <https://doi.org/10.1007/s11886-012-0264-z>
- Marinescu KK, Uriel N, Mann DL, Burkhoff D. Left ventricular assist device-induced reverse remodeling: it's not just about myocardial

- recovery. *Expert Rev Med Devices*. 2017;14:15-26. <https://doi.org/10.1080/17434440.2017.1262762>
22. Blumer V, Kanwar MK, Barnett CF, et al. Cardiogenic shock in older adults: a focus on age-associated risks and approach to management: a scientific statement from the American Heart Association. *Circulation*. 2024;149:e1051-e1065. <https://doi.org/10.1161/cir.0000000000001214>
23. Chioncel O, Parissis J, Mebazaa A, et al. Epidemiology, pathophysiology and contemporary management of cardiogenic shock—a position statement from the Heart Failure Association of the European Society of Cardiology. *Eur J Heart Fail*. 2020;22:1315-1341. <https://doi.org/10.1002/ehf.1922>
24. Sinha SS, Morrow DA, Kapur NK, Kataria R, Roswell RO. 2025 concise clinical guidance: an ACC expert consensus statement on the evaluation and management of cardiogenic shock: a report of the American College of Cardiology Solution Set Oversight Committee. *J Am Coll Cardiol*. 2025;85:1618-1641. <https://doi.org/10.1016/j.jacc.2025.02.018>
25. Patel SM, Berg DD, Bohula EA, et al. Continuum of preshock to classic cardiogenic shock in the Critical Care Cardiology Trials Network registry. *JACC Heart Fail*. 2024;12:1625-1635. <https://doi.org/10.1016/j.jchf.2024.06.009>
26. Jentzer JC, Burstein B, van Diepen S, et al. Defining shock and preshock for mortality risk stratification in cardiac intensive care unit patients. *Circ Heart Fail*. 2021;14:e007678. <https://doi.org/10.1161/circheartfailure.120.007678>
27. Naidu SS, Baran DA, Jentzer JC, et al. SCAI SHOCK stage classification expert consensus update: a review and incorporation of validation studies: this statement was endorsed by the American College of Cardiology (ACC), American College of Emergency Physicians (ACEP), American Heart Association (AHA), European Society of Cardiology (ESC) Association for Acute Cardiovascular Care (ACVC), International Society for Heart and Lung Transplantation (ISHLT), Society of Critical Care Medicine (SCCM), and Society of Thoracic Surgeons (STS) in December 2021. *J Am Coll Cardiol*. 2022;79:933-946. <https://doi.org/10.1016/j.jacc.2022.01.018>
28. Ton VK, Li S, John K, et al. Serial shock severity assessment within 72 hours after diagnosis: a Cardiogenic Shock Working Group report. *J Am Coll Cardiol*. 2024;84:978-990. <https://doi.org/10.1016/j.jacc.2024.04.069>
29. Zweck E, Ton VK, Kanwar M, et al. CSWG-SCAI stages combined with machine learning-based phenotypes for serial risk stratification in cardiogenic shock. *JACC Heart Fail*. 2025;13:102611. <https://doi.org/10.1016/j.jchf.2025.102611>
30. Blumer V, Hanff TC, Gage A, Schrage B, Kanwar MK. Cardiogenic shock teams: past, present, and future directions. *Circ Heart Fail*. 2025;18:e011630. <https://doi.org/10.1161/circheartfailure.124.011630>
31. Kalra S, Ranard LS, Memon S, et al. Risk prediction in cardiogenic shock: current state of knowledge, challenges and opportunities. *J Card Fail*. 2021;27:1099-1110. <https://doi.org/10.1016/j.cardfail.2021.08.003>
32. Baran DA, Billia F, Randhawa V, et al. Consensus statements from the International Society for Heart and Lung Transplantation consensus conference: heart failure-related cardiogenic shock. *J Heart Lung Transplant*. 2024;43:204-216. <https://doi.org/10.1016/j.healun.2023.10.007>
33. Kapur NK, Kanwar M, Sinha SS, et al. Criteria for defining stages of cardiogenic shock severity. *J Am Coll Cardiol*. 2022;80:185-198. <https://doi.org/10.1016/j.jacc.2022.04.049>
34. Basir MB, Lemor A, Gorgis S, et al. Early utilization of mechanical circulatory support in acute myocardial infarction complicated by cardiogenic shock: the National Cardiogenic Shock Initiative. *J Am Heart Assoc*. 2023;12:e031401. <https://doi.org/10.1161/jaha.123.031401>
35. Geller BJ, Sinha SS, Kapur NK, et al. Escalating and de-escalating temporary mechanical circulatory support in cardiogenic shock: a scientific statement from the American Heart Association. *Circulation*. 2022;146:e50-e68. <https://doi.org/10.1161/cir.0000000000001076>
36. Bhimaraj A, Garan AR, Kong Q, et al. Comparison of heart failure cardiogenic shock patients with axillary and femoral intra-aortic balloon pump: Cardiogenic Shock Working Group report. *J Card Fail*. 2026;32:394-405. <https://doi.org/10.1016/j.cardfail.2025.07.020>
37. Pauley E, Lishmanov A, Schumann S, Gala GJ, van Diepen S, Katz JN. Delirium is a robust predictor of morbidity and mortality among critically ill patients treated in the cardiac intensive care unit. 2015. *Am Heart J*. 2015;170:79-86. 86.e71 <https://doi.org/10.1016/j.ahj.2015.04.013>
38. Ibrahim K, McCarthy CP, McCarthy KJ, et al. Delirium in the cardiac intensive care unit. *J Am Heart Assoc*. 7:e008568. doi:10.1161/jaha.118.008568.
39. Appleton RT, Kinsella J, Quasim T. The incidence of intensive care unit-acquired weakness syndromes: a systematic review. *J Intensive Care Soc*. 2015;16:126-136. <https://doi.org/10.1177/1751143714563016>
40. Thiele H, Akin I, Sandri M, de Waha-Thiele S, et al. One-year outcomes after PCI strategies in cardiogenic shock. *N Engl J Med*. 2018;379:1699-1710. <https://doi.org/10.1056/nejmoa1808788>
41. Hochman JS, Sleeper LA, White HD, et al. One-year survival following early revascularization for cardiogenic shock. *JAMA*. 2001;285:190-192. <https://doi.org/10.1001/jama.285.2.190>
42. Gill GS, Sánchez JS, Thandra A, Kanmanthareddy A, Alla VM, Garcia-Garcia HM. Multivessel vs culprit-vessel only percutaneous coronary interventions in acute myocardial infarction and cardiogenic shock: a systematic review and meta-analysis of prospective randomized and retrospective studies. *Eur Heart J Acute Cardiovasc Care*. 2022;11:558-569. <https://doi.org/10.1093/ehjacc/zuac072>
43. Lemor A, Basir MB, Patel K, et al. Multivessel versus culprit-vessel percutaneous coronary intervention in cardiogenic shock. *JACC Cardiovasc Interv*. 2020;13:1171-1178. <https://doi.org/10.1016/j.jcin.2020.03.012>
44. Choi KH, Yang JH, Park TK, et al. Culprit-only versus immediate multivessel percutaneous coronary intervention in patients with acute myocardial infarction complicating advanced cardiogenic shock requiring venoarterial-extracorporeal membrane oxygenation. *J Am Heart Assoc*. 2023;12:e029792. <https://doi.org/10.1161/jaha.123.029792>
45. Bagnall A, Spyridopoulos I. The evidence base for revascularisation of chronic total occlusions. *Curr Cardiol Rev*. 2014;10:88-98. <https://doi.org/10.2174/1573403x10666140331125659>
46. Masiero G, Cardaioli F, Rodinò G, Tarantini G. When to achieve complete revascularization in infarct-related cardiogenic shock. *J Clin Med*. 2022;11. <https://doi.org/10.3390/jcm11113116>
47. van Veelen A, Coerkamp CF, Somsen YBO, et al. Ten-year outcome of recanalization or medical therapy for concomitant chronic total occlusion after myocardial infarction. *J Am Heart Assoc*. 2024;13:e033556. <https://doi.org/10.1161/jaha.123.033556>
48. Panza Julio A, Chrzanowski L, Bonow Robert O. Myocardial viability assessment before surgical revascularization in ischemic cardiomyopathy. *J Am Coll Cardiol*. 2021;78:1068-1077. <https://doi.org/10.1016/j.jacc.2021.07.004>
49. Simard T, Vemulapalli S, Jung RG, et al. Transcatheter edge-to-edge mitral valve repair in patients with severe mitral regurgitation and cardiogenic shock. *J Am Coll Cardiol*. 2022;80:2072-2084. <https://doi.org/10.1016/j.jacc.2022.09.006>
50. Tang GHL, Estevez-Loureiro R, Yu Y, Prillinger JB, Zaid S, Psotka MA. Survival following edge-to-edge transcatheter mitral valve repair in patients with cardiogenic shock: a nationwide analysis. *J Am Heart Assoc*. 2021;10:e019882. <https://doi.org/10.1161/jaha.120.019882>
51. Falasconi G, Melillo F, Pannone L, et al. Use of edge-to-edge percutaneous mitral valve repair for severe mitral regurgitation in cardiogenic shock: a multicenter observational experience (MITRA-SHOCK study). *Catheter Cardiovasc Interv*. 2021;98:e163-e170. <https://doi.org/10.1002/ccd.29683>
52. Dimitriadis K, Soulaïdopoulos S, Pyrryris N, et al. Transcatheter edge-to-edge repair for severe mitral regurgitation in patients with cardiogenic shock: a systematic review and meta-analysis. *J Am Heart Assoc*. 2025;14:e034932. <https://doi.org/10.1161/jaha.124.034932>
53. Kolte D, Khera S, Vemulapalli S, et al. Outcomes following urgent/emergent transcatheter aortic valve replacement: insights from the STS/ACC TVT registry. *JACC Cardiovasc Interv*. 2018;11:1175-1185. <https://doi.org/10.1016/j.jcin.2018.03.002>
54. Frerker C, Schewel J, Schlüter M, et al. Emergency transcatheter aortic valve replacement in patients with cardiogenic shock due to acutely decompensated aortic stenosis. *Euro-Intervention*. 2016;11:1530-1536. https://doi.org/10.4244/eijy15m03_03
55. Nair RM, Chawla S, Abdelghaffar B, et al. Comparison of contemporary treatment strategies

- in patients with cardiogenic shock due to severe aortic stenosis. *J Am Heart Assoc.* 2024;13:e033601. <https://doi.org/10.1161/jaha.123.033601>
56. Debry N, Kone P, Vincent F, et al. Urgent balloon aortic valvuloplasty in patients with cardiogenic shock related to severe aortic stenosis: time matters. *EuroIntervention.* 2018;14:e519-e525. <https://doi.org/10.4244/eij-d-18-00029>
 57. Carnicelli AP, Miller PE, Berg DD, et al. Characteristics and outcomes of patients with cardiogenic shock and clinically significant valvular heart disease: from the Critical Care Cardiology Trials Network. *J Card Fail.* <https://doi.org/10.1016/j.cardfail.2025.01.019>.
 58. Chiang CJ, Kerolos M, Sunnaa M, et al. Investigation of outcomes following transcatheter edge to edge repair of the mitral valve versus medical management alone in patients with cardiogenic shock and mitral regurgitation. *Am Heart J Plus.* 2024;45:100430. <https://doi.org/10.1016/j.ahj.2024.100430>
 59. Zhong J, Kamp N, Bansal A, et al. Balloon aortic valvuloplasty in the modern era: a review of outcomes, indications, and technical advances. *J Soc Cardiovasc Angiogr Interv.* 2023;2:101002. <https://doi.org/10.1016/j.jscai.2023.101002>
 60. Ballout JA, Wazni OM, Tarakji KG, et al. Catheter ablation in patients with cardiogenic shock and refractory ventricular tachycardia. *Circ Arrhythm Electrophysiol.* 2020;13:e007669. <https://doi.org/10.1161/circep.119.007669>
 61. Baratto F, Pappalardo F, Oloriz T, et al. Extracorporeal membrane oxygenation for hemodynamic support of ventricular tachycardia ablation. *Circ Arrhythm Electrophysiol.* 2016;9. <https://doi.org/10.1161/circep.116.004492>
 62. Al-Shakarchi NJ, Ho JSY, Bray JJH, et al. Cardiac resynchronization therapy in inotrope-dependent heart failure: a meta-analysis. *ESC Heart Fail.* 2024;11:2616-2626. <https://doi.org/10.1002/ehf2.14835>
 63. Bhattacharya S, Abebe K, Simon M, Saba S, Adelstein E. Role of cardiac resynchronization in end-stage heart failure patients requiring inotrope therapy. *J Card Fail.* 2010;16:931-937. <https://doi.org/10.1016/j.cardfail.2010.07.253>
 64. Spartalis M, Zweiker D, Spartalis E, Iliopoulos DC, Siasos G. Hemodynamic support during catheter ablation of ventricular arrhythmias in patients with cardiogenic shock. *Front Cardiovasc Med.* 2023;10:1145123. <https://doi.org/10.3389/fcvm.2023.1145123>
 65. Dusik M, Havranek S, Rob D, et al. Catheter ablation in patients on mechanical circulatory supports for cardiogenic shock. *PLoS One.* 2025;20:e032597. <https://doi.org/10.1371/journal.pone.032597>
 66. Hernandez GA, Blumer V, Arcay L, et al. Cardiac Resynchronization therapy in inotrope-dependent heart failure patients: a systematic review and meta-analysis. *JACC Heart Fail.* 2018;6:734-742. <https://doi.org/10.1016/j.jchf.2018.02.016>
 67. Bandini M, D'Ettore N, Iannotti W, et al. Midterm outcomes of patients with native heart recovery after Impella 5+ for cardiogenic shock. *Eur J Heart Fail.* 2025;27:3336-3341. <https://doi.org/10.1002/ehj.3544>
 68. Deschamps J, Obanor O, Clarizia N, et al. Best management practices on temporary mechanical circulatory support: joint consensus report of the Perioperative Quality Initiative and the Enhanced Recovery After Surgery Cardiac Society. *Ann Thorac Surg.* 2025;120:225-243. <https://doi.org/10.1016/j.athoracsur.2025.01.039>
 69. Yokawa J, Nishikawa T, Shimojima M, et al. Early initiation of guideline-directed medical therapy in patients with cardiogenic shock supported with Impella: a two-center retrospective study. *BMC Cardiovasc Disord.* 2025;25:818. <https://doi.org/10.1186/s12872-025-05311-5>
 70. Bernhardt AM, Blumer V, Vandenbriele C, et al. Clinical management of the Impella 5.5 pump. *J Heart Lung Transplant.* 2025;44:1688-1702. <https://doi.org/10.1016/j.healun.2025.06.008>
 71. McNamara DM, Starling RC, Cooper LT, et al. Clinical and demographic predictors of outcomes in recent onset dilated cardiomyopathy: results of the IMAC (Intervention in Myocarditis and Acute Cardiomyopathy)-2 study. *J Am Coll Cardiol.* 2011;58:1112-1118. <https://doi.org/10.1016/j.jacc.2011.05.033>
 72. Fried J, Farr M, Kanwar M, et al. Clinical outcomes among cardiogenic shock patients supported with high-capacity Impella axial flow pumps: a report from the Cardiogenic Shock Working Group. *J Heart Lung Transplant.* 2024;43:1478-1488. <https://doi.org/10.1016/j.healun.2024.05.015>
 73. Baritussio A, Scatteia A, Bucciarelli-Ducci C. Role of cardiovascular magnetic resonance in acute and chronic ischemic heart disease. *Int J Cardiovasc Imaging.* 2018;34:67-80. <https://doi.org/10.1007/s10554-017-1116-0>
 74. Jozwiak M, Lim SY, Si X, Monnet X. Biomarkers in cardiogenic shock: old pals, new friends. *Ann Intensive Care.* 2024;14:157. <https://doi.org/10.1186/s13613-024-01388-x>
 75. Landsteiner I, Ikoma T, Lewis GD. Cardiopulmonary exercise testing in advanced heart failure management. *Heart Fail Clin.* 2025;21:35-49. <https://doi.org/10.1016/j.hfc.2024.09.001>
 76. Mazzarotto F, Tayal U, Buchan RJ, et al. Reevaluating the genetic contribution of monogenic dilated cardiomyopathy. *Circulation.* 2020;141:387-398. <https://doi.org/10.1161/circulationaha.119.037661>
 77. Bui QM, Ding J, Hong KN, Adler EA. The genetic evaluation of dilated cardiomyopathy. *Struct Heart.* 2023;7:100200. <https://doi.org/10.1016/j.shj.2023.100200>
 78. Lee C, Tully A, Fang JC, et al. Building and optimizing the interdisciplinary heart team. *J Soc Cardiovasc Angiogr Interv.* 2023;2:101067. <https://doi.org/10.1016/j.jscai.2023.101067>
 79. Maury P, Mansourati J, Fauchier L, Waintraub X, Boveda S, Sacher F. Management of sustained arrhythmias for patients with cardiogenic shock in intensive cardiac care units. *Arch Cardiovasc Dis.* 2019;112:781-791. <https://doi.org/10.1016/j.acvd.2019.10.002>
 80. Dimond MG, Rosner CM, Lee SB, et al. Guideline-directed medical therapy implementation during hospitalization for cardiogenic shock. *ESC Heart Fail.* 2025;12:60-70. <https://doi.org/10.1002/ehf2.14863>
 81. Isseh IN, McBurnie A, Aziz AAA, et al. The cardiogenic shock recovery clinic. *J Card Fail.* Published online November 2, 2025. <https://doi.org/10.1016/j.cardfail.2025.10.009>
 82. Blumer V, Januzzi JL Jr, Lindenfeld J, et al. Heart failure drug development over the eras: from the Heart Failure Collaboratory. *JACC Heart Fail.* 2024;12:1803-1813. <https://doi.org/10.1016/j.jchf.2024.03.021>
 83. Bakhr RN, Davidson JF, Bookstaver RE, et al. Implementation of an ICU recovery clinic at a tertiary care academic center. *Crit Care Explor.* 2019;1:e0034. <https://doi.org/10.1097/ccx.000000000000034>
 84. Schwitzer E, Jensen KS, Brinkman L, et al. Survival ≠ recovery: a narrative review of post-intensive care syndrome. *CHEST Crit Care.* 2023;1:100003. <https://doi.org/10.1016/j.chstcc.2023.100003>
 85. Doherty DA, Tong SYC, Reilly J, et al. Registry randomised trials: a methodological perspective. *BMJ Open.* 2023;13:e068057. <https://doi.org/10.1136/bmjopen-2022-068057>
 86. Buccheri S, Sarno G, Fröbert O, et al. Assessing the nationwide impact of a registry-based randomized clinical trial on cardiovascular practice. *Circ Cardiovasc Interv.* 2019;12:e007381. <https://doi.org/10.1161/circinterventions.118.007381>
 87. Waksman R, Pahuja M, Diepen SV, et al. Standardized definitions for cardiogenic shock research and mechanical circulatory support devices: scientific expert panel from the Shock Academic Research Consortium (SHARC). *Circulation.* 2023;148:1113-1126. <https://doi.org/10.1161/circulationaha.123.064527>
 88. Spertus JA, Jones PG, Sandhu AT, Arnold SV. Interpreting the Kansas City Cardiomyopathy Questionnaire in clinical trials and clinical care: JACC state of the art review. *J Am Coll Cardiol.* 2020;76:2379-2390. <https://doi.org/10.1016/j.jacc.2020.09.542>
 89. Butler J, Usman MS, Gandotra C, et al. Patient-reported outcomes as endpoints in heart failure trials. *Circulation.* 2025;151:1111-1125. <https://doi.org/10.1161/circulationaha.124.072158>
 90. Patel SM, Lopes MS, Morrow DA, et al. Targeted proteomic profiling of cardiogenic shock in the cardiac intensive care unit. *Eur Heart J Acute Cardiovasc Care.* 2024;13:624-628. <https://doi.org/10.1093/ehjacc/zaee068>

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